

Geometry of the Two-Dimensional Sphere $\mathcal{S}^2$

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Context

The conceptual structure of this derivation follows material introduced in advanced coursework at the Universitat de Barcelona. All detailed calculations, formal development, and typesetting are the independent work of the author.

Introduction

Let us consider the sphere $\mathcal{S}^2$ of constant radius a . The metric of this two-dimensional curved space in standard spherical coordinates is given by the line element $ds^2 = a^2(d\theta^2 + \sin^2\theta d\phi^2)$. In this document, we will derive the fundamental geometric tensors of this manifold.

1. The Christoffel Symbols via Euler-Lagrange

To calculate the Christoffel symbols efficiently, we utilize the fact that, in a curved space without external forces, the mechanical trajectory of a free particle is a geodesic. This trajectory satisfies $\ddot{x}^\lambda + \Gamma_{\mu\nu}^\lambda \dot{x}^\mu \dot{x}^\nu = 0$. For the sphere, utilizing the coordinates $x^\mu = (\theta, \phi)$, we have:

$$\begin{aligned}\ddot{\theta} + \Gamma_{\theta\theta}^\theta \dot{\theta}^2 + \Gamma_{\theta\phi}^\theta \dot{\theta} \dot{\phi} + \Gamma_{\phi\theta}^\theta \dot{\phi} \dot{\theta} + \Gamma_{\phi\phi}^\theta \dot{\phi}^2 &= \ddot{\theta} + \Gamma_{\theta\theta}^\theta \dot{\theta}^2 + 2\Gamma_{\theta\phi}^\theta \dot{\theta} \dot{\phi} + \Gamma_{\phi\phi}^\theta \dot{\phi}^2 = 0 \\ \ddot{\phi} + \Gamma_{\theta\theta}^\phi \dot{\theta}^2 + \Gamma_{\theta\phi}^\phi \dot{\theta} \dot{\phi} + \Gamma_{\phi\theta}^\phi \dot{\phi} \dot{\theta} + \Gamma_{\phi\phi}^\phi \dot{\phi}^2 &= \ddot{\phi} + \Gamma_{\theta\theta}^\phi \dot{\theta}^2 + 2\Gamma_{\theta\phi}^\phi \dot{\theta} \dot{\phi} + \Gamma_{\phi\phi}^\phi \dot{\phi}^2 = 0\end{aligned}$$

These equations can be obtained via the principle of least action using a Lagrangian of the form $\mathcal{L} = g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu$. First, we must determine the metric of the sphere. Let:

$$\begin{aligned}\psi : \Omega &\longrightarrow \mathbb{R}^3 \\ (\theta, \phi) &\longmapsto (a \sin \theta \cos \phi, a \sin \theta \sin \phi, a \cos \theta)\end{aligned}$$

be a local chart of the sphere $\mathcal{S}^2$ where $\Omega = (0, \pi) \times (0, 2\pi) \subset \mathbb{R}^2$ is an open set. The space spanned by $\psi_\theta := \frac{\partial \psi}{\partial \theta}$ and $\psi_\phi := \frac{\partial \psi}{\partial \phi}$ is the tangent space $T_p \psi = T_q \mathcal{S}^2$, $\forall q = \psi(p)$. Note that the points $\{0, \pi\} \times \{0, 2\pi\} \notin \Omega$, so the formalism presented here does not apply at the coordinate singularities. Since ψ_θ and ψ_ϕ form a basis of $T_q \mathcal{S}^2$, we know that $\forall v \in T_p \mathcal{S}^2$, $\exists \lambda, \mu : \Omega \rightarrow \mathbb{R}$ such that $v = \lambda \psi_\theta + \mu \psi_\phi$. Consequently, $\forall \alpha, \beta \in T_q \mathcal{S}^2$, the scalar product is given by:

$$\begin{aligned}\langle \alpha, \beta \rangle &= \langle \lambda \psi_\theta + \mu \psi_\phi, \rho \psi_\theta + \sigma \psi_\phi \rangle \\ &= \lambda \rho \langle \psi_\theta, \psi_\theta \rangle + \lambda \sigma \langle \psi_\theta, \psi_\phi \rangle + \mu \rho \langle \psi_\phi, \psi_\theta \rangle + \mu \sigma \langle \psi_\phi, \psi_\phi \rangle \\ &= \begin{pmatrix} \lambda & \mu \end{pmatrix} \begin{pmatrix} \langle \psi_\theta, \psi_\theta \rangle & \langle \psi_\theta, \psi_\phi \rangle \\ \langle \psi_\phi, \psi_\theta \rangle & \langle \psi_\phi, \psi_\phi \rangle \end{pmatrix} \begin{pmatrix} \rho \\ \sigma \end{pmatrix} = g_{\mu\nu} \alpha^\mu \beta^\nu\end{aligned}$$

where $g_{\mu\nu}$ is the metric of the manifold, permitting geometry via the coordinate space. In the specific case of the sphere, the tangent basis vectors are:

$$\psi_\theta = (a \cos \theta \cos \phi, a \cos \theta \sin \phi, -a \sin \theta), \quad \psi_\phi = (-a \sin \theta \sin \phi, a \sin \theta \cos \phi, 0)$$

Evaluating their respective inner products yields the metric components:

$$g_{\theta\theta} = \langle \psi_\theta, \psi_\theta \rangle = a^2, \quad g_{\phi\phi} = \langle \psi_\phi, \psi_\phi \rangle = a^2 \sin^2 \theta, \quad g_{\theta\phi} = g_{\phi\theta} = \langle \psi_\theta, \psi_\phi \rangle = 0$$

After this brief topological parenthesis, we can calculate the equations of motion using the Lagrangian $\mathcal{L} = a^2 \dot{\theta}^2 + a^2 \sin^2 \theta \dot{\phi}^2$:

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}} - \frac{\partial \mathcal{L}}{\partial \theta} = 2a^2 \ddot{\theta} - 2a^2 \sin \theta \cos \theta \dot{\phi}^2 = 0 \quad \implies \quad \ddot{\theta} - \sin \theta \cos \theta \dot{\phi}^2 = 0$$

$$\begin{aligned} \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\phi}} - \frac{\partial \mathcal{L}}{\partial \phi} &= 2a^2 \frac{d}{dt} (\sin^2 \theta \dot{\phi}) = 0 \\ &= 2a^2 (2 \sin \theta \cos \theta \dot{\theta} \dot{\phi} + \sin^2 \theta \ddot{\phi}) = 0 \quad \implies \quad \ddot{\phi} + 2 \cot \theta \dot{\theta} \dot{\phi} = 0 \end{aligned}$$

From here, comparing these results with the general geodesic expressions:

$$\Gamma_{\phi\phi}^\theta = -\sin \theta \cos \theta, \quad \Gamma_{\phi\theta}^\phi = \cot \theta$$

2. The Riemann Curvature Tensor

Because $\mathcal{S}^2$ is a two-dimensional curved space, the number of independent components of the Riemann tensor is strictly one. We evaluate the non-vanishing mixed component $R_{\phi\theta\phi}^\theta$ using the standard definition:

$$\begin{aligned} R_{\sigma\mu\nu}^\rho &= \partial_\mu \Gamma_{\nu\sigma}^\rho - \partial_\nu \Gamma_{\mu\sigma}^\rho + \Gamma_{\mu\lambda}^\rho \Gamma_{\nu\sigma}^\lambda - \Gamma_{\nu\lambda}^\rho \Gamma_{\mu\sigma}^\lambda \\ \implies R_{\phi\theta\phi}^\theta &= \partial_\theta \Gamma_{\phi\phi}^\theta - \cancel{\partial_\phi \Gamma_{\theta\phi}^\theta} + \Gamma_{\theta\lambda}^\theta \Gamma_{\phi\phi}^\lambda - \Gamma_{\phi\lambda}^\theta \Gamma_{\theta\phi}^\lambda \\ &= \partial_\theta \Gamma_{\phi\phi}^\theta + \cancel{\Gamma_{\theta\theta}^\theta \Gamma_{\phi\phi}^\theta} + \Gamma_{\theta\phi}^\theta \Gamma_{\phi\phi}^\phi - \Gamma_{\phi\theta}^\theta \Gamma_{\theta\phi}^\phi - \Gamma_{\phi\phi}^\theta \Gamma_{\theta\phi}^\phi \\ &= \partial_\theta (-\sin \theta \cos \theta) + (\sin \theta \cos \theta)(\cot \theta) \\ &= -(\cos^2 \theta - \sin^2 \theta) + \cos^2 \theta = \sin^2 \theta \\ \implies R_{\phi\theta\phi}^\theta &= g_{\theta\theta} R_{\phi\theta\phi}^\theta = a^2 \sin^2 \theta \end{aligned}$$

We deduce that the sphere is a curved space because this component is non-zero for any value of θ , excluding the coordinate singularities at $\theta = 0, \pi$. As discussed, the differential geometric formalism relies on the metric defined within a specific local chart. The complete polar atlas of the sphere requires at least two overlapping charts to cover the manifold; the vanishing of the tensor at the poles in this representation is merely an artifact of the coordinate choice, not a reflection of local flatness.

3. The Ricci Tensor and Scalar Curvature

We can explicitly calculate the components using the definition of the Ricci tensor $R_{\mu\nu} = g^{\alpha\beta} R_{\alpha\mu\beta\nu}$. We first demonstrate that the Ricci tensor is symmetric:

$$R_{\nu\mu} = g^{\alpha\beta} R_{\alpha\nu\beta\mu} \stackrel{(1)}{=} g^{\beta\alpha} R_{\beta\nu\alpha\mu} \stackrel{(2)}{=} g^{\beta\alpha} R_{\alpha\mu\beta\nu} \stackrel{(3)}{=} g^{\alpha\beta} R_{\alpha\mu\beta\nu} = R_{\mu\nu}$$

where in (1) the dummy indices are renamed, in (2) the symmetry $R_{\{\alpha\mu,\beta\nu\}}$ is applied, and in (3) the symmetry of the metric tensor is used. The components of the Ricci tensor are (considering $g^{\theta\phi} = g^{\phi\theta} = 0$):

$$R_{\theta\theta} = g^{\alpha\beta} R_{\alpha\theta\beta\theta} = g^{\theta\theta} R_{\theta\theta\theta\theta} + g^{\phi\phi} R_{\phi\theta\phi\theta} = \frac{1}{a^2 \sin^2 \theta} a^2 \sin^2 \theta = 1$$

$$R_{\phi\phi} = g^{\alpha\beta} R_{\alpha\phi\beta\phi} = g^{\theta\theta} R_{\theta\phi\theta\phi} + g^{\phi\phi} R_{\phi\phi\phi\phi} = \frac{1}{a^2} a^2 \sin^2 \theta = \sin^2 \theta$$

$$R_{\phi\theta} = R_{\theta\phi} = g^{\alpha\beta} R_{\alpha\theta\beta\phi} = g^{\theta\theta} R_{\theta\theta\theta\phi} + g^{\phi\phi} R_{\phi\theta\phi\phi} = 0$$

With these components, the Ricci scalar is:

$$R = g^{\mu\nu} R_{\mu\nu} = g^{\theta\theta} R_{\theta\theta} + g^{\phi\phi} R_{\phi\phi} = \frac{1}{a^2} (1) + \frac{1}{a^2 \sin^2 \theta} (\sin^2 \theta) = \frac{2}{a^2}$$

4. The Einstein Tensor

From the definition $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R$, we can compute the tensor directly in matrix form:

$$\begin{aligned} G_{\mu\nu} &= \begin{pmatrix} R_{\theta\theta} - \frac{1}{2} g_{\theta\theta} R & R_{\theta\phi} - \frac{1}{2} g_{\theta\phi} R \\ R_{\phi\theta} - \frac{1}{2} g_{\phi\theta} R & R_{\phi\phi} - \frac{1}{2} g_{\phi\phi} R \end{pmatrix} \\ &= \begin{pmatrix} 1 - \frac{1}{2} a^2 \left(\frac{2}{a^2} \right) & 0 \\ 0 & \sin^2 \theta - \frac{1}{2} a^2 \sin^2 \theta \left(\frac{2}{a^2} \right) \end{pmatrix} \end{aligned}$$

The Einstein Tensor of S^2

The resulting Einstein tensor vanishes identically:

$$G_{\mu\nu} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

This confirms the fundamental property that the Einstein tensor is identically zero for any manifold of dimension $n \leq 2$.